Qatar-2b

R-1719

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-2_190501` · created 2026-08-02T16:18:49+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458604.70123 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.213 +/- 0.011
Transit depth (Rp/Rs)^2	4.54 +/- 0.45 [%]
Orbital Inclination (inc)	89.78 +/- 1.18
Airmass coefficient 1 (a1)	1.268 +/- 0.031
Airmass coefficient 2 (a2)	-0.1746 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	2.48 %
Best Comparison Star	None
Optimal Aperture	4.12
Optimal Annulus	9.74
Transit Duration (day)	0.08005 +/- 0.00098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.213 ± 0.011 vs 0.16327 (deviation 4.21σ)
Duration fit vs published	0.08005 d vs 0.0767 d (deviation 3.22σ)
Mid-time O–C	2.0 min
Depth SNR	18.0
Residual scatter	2.48 %

Light curve

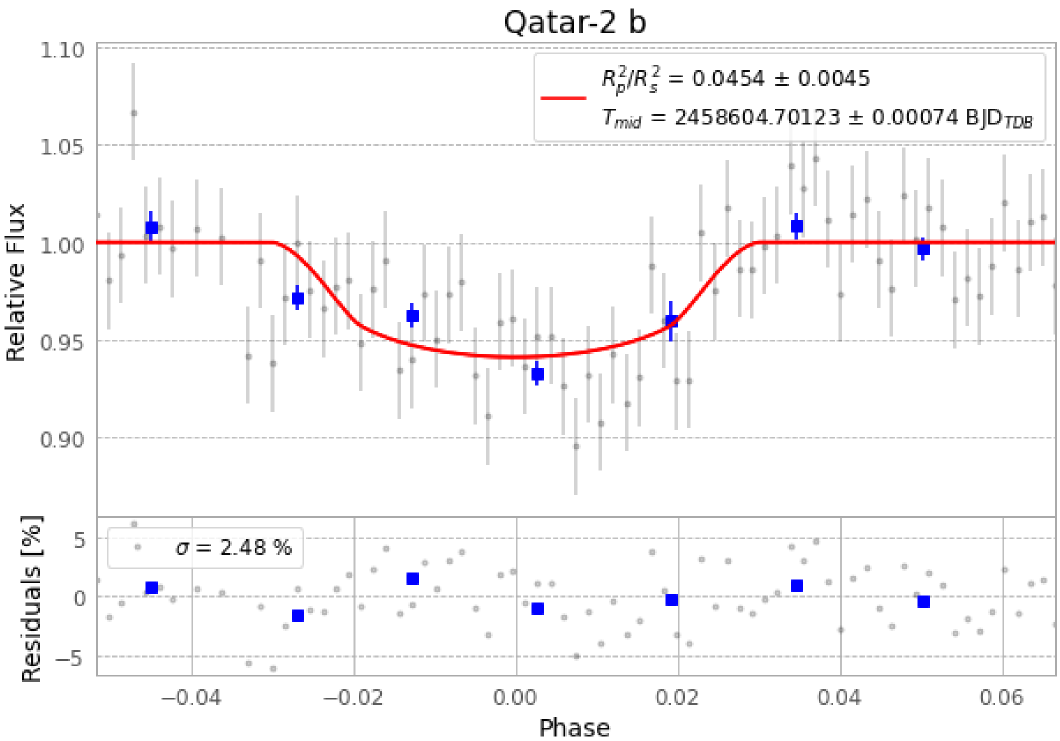

Figure 1 — FinalLightCurve_Qatar-2 b_2019-04-30.png (SHA-256 2f698af624e82ef4…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [470, 32], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2ef8fee808c79187…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `Qatar-2190501030012.FITS` → `Qatar-2190501064812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-190501.log (e3243ce53e76…), FinalLightCurve_Qatar-2 b_2019-04-30.png (2f698af624e8…), FinalLightCurve_Qatar-2 b_2019-04-30.csv (2e0ec2d735fd…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

 Generated 2026-08-02 16:18Z.